

Developmental Learning Architecture: Learning History, Fast-Weight Traces, and Future Adaptation

Draft v0.2 (validation-stage) — workshop/arXiv

Abstract

We study whether a learner's prior learning history changes how it will learn in the future, and which internal state carries this effect. Our Developmental Learning Architecture (DLA) keeps standard MLP/Transformer backbones unchanged but augments every weight with per-parameter fast weights, plasticity, and sleep consolidation, forming a fast/slow learner state. Using a 6.59M Chinese character-level GPT we show: (i) DLA matches tuned AdamW on new-domain adaptation while keeping consolidated slow memory essentially immune to forgetting; (ii) different curriculum histories (easy->hard vs hard->easy) lead to different future adaptation on an unseen domain; (iii) this history-dependent future adaptation is causally tied to the fast-weight state w_{fast} — replacing the hard->easy learner's fast weights with those from an easy->hard learner significantly degrades future adaptation ($n=12$, gain@40 difference -0.019, bootstrap CI excludes zero, Cohen's $d \sim -0.7$); (iv) norm-matching and within-matrix shuffling do not remove the effect, so it is not explained by global magnitude or by a simple random-parameter control; (v) localization is suggestive of MLP and attention contributions; (vi) standard baselines (AdamW, replay, EWC) also show history sensitivity, but DLA is the most robust after a difficult history; and (vii) the destructive w_{fast} effect replicates on a second Shakespeare character-level backbone. We do not find evidence that more experience monotonically accelerates learning: controlled longitudinal experiments across same-domain, cross-domain, and physics-near-transfer settings consistently fail to show a positive trend. The paper therefore proposes a precise framing: developmental change does not equal universally faster learning; rather, learning history leaves a persistent, partially localizable trace in transient learner state, with w_{fast} providing a causal component of history-dependent future adaptation.

1 Introduction

Standard neural training assumes the learner is a fixed function plus an external optimizer. This paper asks a more developmental question: can the way a model learns change as a result of what it has learned? We build a minimal architecture, DLA, in which each weight matrix is augmented by fast weights, plasticity, and sleep consolidation. The Transformer backbone itself is not modified.

Our empirical questions are:

- Does learning history change future adaptation on a new task?
- Which internal state carries that effect?
- Does the effect manifest as faster learning (monotonic self-improvement)?

We find support for the first two, and a clear negative result for the third.

2 Related Work

- Fast weights and fast/slow memory (Hinton & Plaut, 1987; Ba et al., 2016; Schmidhuber, 1992).
- Stability-plasticity dilemma and synaptic intelligence (Abraham & Robins, 2005; Zenke et al., 2017).
- Meta-learning and learned plasticity (Andrychowicz et al., 2016; Miconi et al., 2019; Beaulieu et al., 2020).
- Continual learning baselines used here: Elastic Weight Consolidation (Kirkpatrick et al., 2017) and experience replay (Robins, 1995).
- This work is distinct: we do not primarily propose a better continual-learning algorithm; we ask a mechanistic question about whether and where "development" lives in the learner state.

3 Method

3.1 DLA state

For every 2D weight matrix (or embedding) we maintain:

- w_{slow} : consolidated long-term knowledge;
- w_{fast} : current learning trace;
- P : per-parameter plasticity gate;

● Q: slow-eligibility trace.

Effective weights:

$W_{\text{eff}} = W_{\text{slow}} + \text{softplus}(P) * W_{\text{fast}}$

The online (wake) update uses Adam-style moments on W_{fast} , modulated by $\text{softplus}(P)$, with P updated by a meta-plasticity rule. At task boundaries (sleep), W_{slow} is updated from Q and a small direct transfer of W_{fast} , and W_{fast} is decayed.

3.2 Histories and probe task

To create two different developmental histories we use three curriculum domains:

- easy->hard (EH): domains ordered from low birth perplexity to high;
- hard->easy (HE): the reverse order.

After experiencing all three tasks, the same model (stateful DLA) is probed on an unseen domain D under a fixed adaptation budget. We measure test perplexity trajectories and report gain@40 (relative PPL reduction after 40 steps), normalized learning efficiency, and T80.

4 Experimental Setup

Primary backbone: - 6.59M Chinese character GPT (6 layers, 8 heads, 256 dim, vocab 7280), pretrained on Chinese Wikipedia. - Curriculum domains: Wikipedia, SFT-style QA, science Wikipedia. - Unseen D : science slice not used in curriculum.

Second setting: - ~10.65M Shakespeare character GPT (6 layers, 6 heads, 384 dim, vocab 65).

Baselines: - AdamW; AdamW + replay; EWC, all using the same histories and D probe where applicable.

Metrics: - gain@40; LE $_D$; T80; initial PPL; per-task normalized loss slope; paired contrasts with bootstrap CI and sign tests.

5 Results

5.1 Fast/slow separation protects consolidated memory

Stage 4 formal (5 seeds): DLA matches tuned AdamW on adaptation to a new domain (gain +14.0% vs +13.0%) while consolidated slow-memory forgetting is ~0 (-0.4%/-0.2). This shows the fast/slow separation is a usable memory mechanism.

5.2 Learning history changes future adaptation

Across several settings, a hard->easy history leads to better future adaptation on unseen D than easy->hard. This history effect is not unique to DLA; standard learners also show it (Section 5.6), but DLA is the most robust after a difficult history.

5.3 Scalar learning-rule parameters are not a stable carrier

Body $\times$ phi cross-injection (Stage 5.5c): replacing the 9-dimensional tempo vector between histories changes future adaptation much less than replacing the body state. We therefore say phi is not a stable/dominant carrier in this setting, not that phi has no effect.

5.4 W_{fast} is a causal component

The core causal result (Stage 5.5e P0, $n=12$):

condition	mean gain@40
HE/HE	+0.014
HE + EH W_{fast}	-0.005

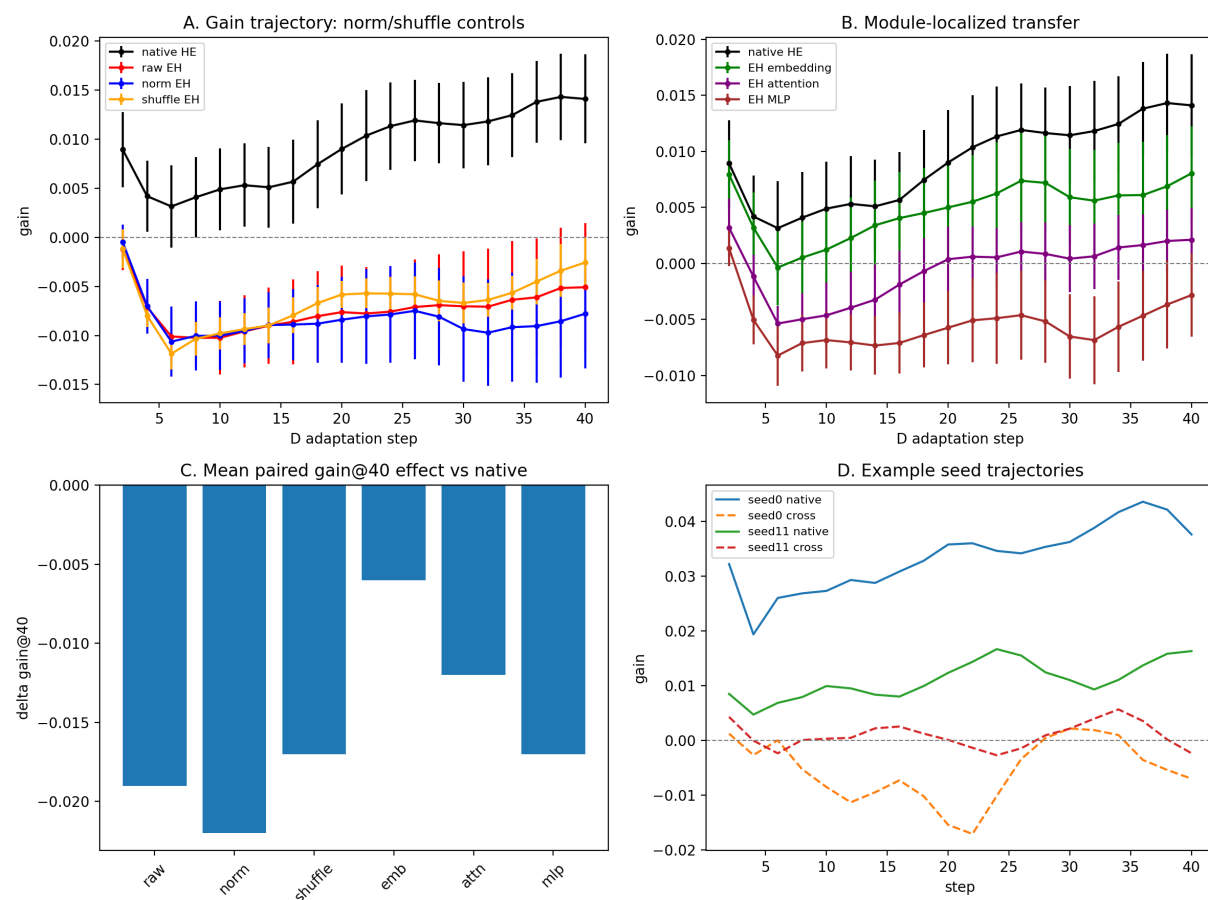
Paired contrast HE->raw EH: mean -0.019, 95% CI [-0.033, -0.005], Cohen's $d \sim -0.71$, sign $p=0.019$ (10/12 negative). Initial PPL differences are small; the effect appears in the adaptation trajectory.

5.5 Norm, shuffle and module controls

n=12 gain@40 paired contrasts (vs HE/HE):

control	mean delta gain@40	95% CI	d
raw EH W_fast	-0.019	[-0.034, -0.005]	-0.71
norm-matched EH W_fast	-0.022	[-0.034, -0.009]	-0.90
shuffled EH W_fast	-0.017	[-0.023, -0.009]	-1.25
EH embedding W_fast	-0.006	[-0.009, -0.002]	-0.87
EH attention W_fast	-0.012	[-0.018, -0.006]	-1.06
EH MLP W_fast	-0.017	[-0.025, -0.008]	-1.05

Norm-matching does not rescue the effect; global magnitude is not the explanation. Shuffle also does not rescue the effect, so a simple random-parameter control is not sufficient. Module localization is suggestive of MLP and attention contributions; embedding contributes less.



5.6 Fair baselines

n=5 (LE_D means):

method	EH	HE	HE-EH
AdamW	0.339	0.675	+0.336
AdamW + replay	0.375	0.961	+0.586
EWC	0.292	0.895	+0.603
DLA	0.517	0.709	+0.192

DLA is the most robust after an EH (easy->hard) history. Standard learners achieve higher HE performance but suffer larger EH degradation.

5.7 Second-setting replication

Shakespeare character GPT, 2 seeds:

	seed	HE/HE gain40	HE + EH W_fast
0		+0.052	+0.004
1		+0.042	-0.001

The destructive W_fast effect replicates in direction on a different backbone/corpus. Small n; treated as exploratory support.

5.8 Negative result: development $\neq$ monotonically faster learning

We directly tested whether more experience makes later tasks easier to learn:

- Stage 5 (difficulty-normalized curriculum): LE roughly flat.
- Stage 6 (matched-difficulty science slices): 7/10 seeds directional, $p=0.17$.
- Stage 7 (cross-domain, 10 tasks, 20 seeds): regression slope ~ 0 , $p=0.82$.
- Stage 8 (physics near-transfer, 20 seeds): $d=-0.17$.

We therefore do not claim that developmental learning produces universally faster learning. The evidence supports a more specific statement: learning history changes the learner's state and future adaptation, and this change is not simply "getting faster."

6 Discussion

- What is safe to claim. DLA separates fast adaptation from slow memory preservation. Learning history changes future adaptation. The effect is substantially body-mediated, with W_fast as a causal component. Norm and shuffle controls do not remove the effect. The effect is not monotonically faster learning.
- What remains suggestive. Exact localization to one module; the degree to which W_fast encodes content vs dynamics; generalizability beyond two small backbones.
- What must not be claimed. That W_fast is the only carrier; that phi has no effect; that development means universally faster learning; that sleep consolidation strongly writes knowledge into W_s Low; that results at 6–10M parameters transfer to large language models.
- Reviewer-facing limitations. No pre-registration (we report $n=10 \rightarrow n=12$ extension transparently). Baselines $n=5$. Second setting $n=2$. One language plus Shakespeare English corpus only.

7 Conclusion

We provide evidence for the claim that a learner is not fully characterized by what it currently knows: its prior learning history can alter how it subsequently learns. The evidence points to a persistent, partially localizable trace in transient learner state, with W_fast providing a causal component of history-dependent future adaptation. We do not find evidence that this developmental effect simply manifests as faster learning.

Reproducibility

- Code: <https://github.com/JayCRL/DLA>
- Experiment tables and result JSONs: see docs/experiments.md, paper/overnight_report.md, paper/validation_audit.md, results/.
- Scripts:
 - stage4_formal.py, stage5_progressive_curriculum.py
 - stage5e_wfast_p0.py, validation_p0_controls.py, validation_baselines.py, validation_second_setting.py
 - stage6_longitudinal.py, stage7_cross_domain.py, stage8_physics.py
- Figures: see paper/figures/.

References (to be completed)

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